

Representation Inversion Theory

A Philosophical Position on Mathematical Work under Machine-Scale Derivation

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Abstract

Artificial intelligence is beginning to alter a basic condition under which mathematics has historically been practiced. Mathematical derivations, formal objects, candidate representations, searches, and proofs can now be produced at scales that humans cannot inspect in the form in which they were generated. The resulting problem is not simply whether machines can prove theorems. It is how human mathematical work should be organized once the production of mathematical structure exceeds the capacity of humans to assimilate that structure directly.

Representation Inversion Theory (RIT) is a philosophical response to this change. Its central claim is simple: *when machine capacity overwhelms a level of mathematical representation, humans should not be required to compete with machines at that level. Mathematical work should move to a representation appropriate to the remaining human tasks.*

RIT does not claim that abstraction, modularity, interfaces, specialization, formal verification, or mathematical division of labour are new. Mathematics has always depended on them. Nor does it propose a new foundation for mathematics. The claim is instead that machine-scale derivation changes their status. Methods previously used to manage difficult mathematics may increasingly become necessary principles for organizing mathematical labour.

This paper develops only the philosophical part of a broader RIT research programme. Companion work is intended to examine the epistemic and economic consequences of representation inversion in real mathematical practice, its concrete mathematical realization, and the complexity-theoretic conditions under which further inversions become necessary.

1 The problem is no longer only proof production

For most of mathematical history, difficult derivations were expensive. Producing a proof, exploring many candidate constructions, searching a large space of representations, or formalizing a complicated argument demanded substantial human effort. Human mathematical attention and mathematical production therefore operated on roughly compatible scales.

Machine reasoning changes this relation. A sufficiently capable automated system can search, derive, compare, formalize, reject, and regenerate mathematical structures at a scale that no individual mathematician, and eventually no mathematical community, could reasonably reconstruct step by step.

This creates a new asymmetry:

$$\text{machine derivational capacity} > \text{human assimilative capacity}.$$

The inequality is conceptual rather than numerical. It describes a situation in which the amount of mathematical activity that can be produced or searched becomes substantially larger than the amount that humans can keep cognitively active.

This pressure is already visible in contemporary discussions of mathematics and artificial intelligence. Tao describes a possible transition from *proof scarcity* to *proof abundance*, in which proof generation, verification, exposition, digestion, and canonicalization cease to occur at comparable rates [8]. Venkatesh similarly argues that reasoning machines will affect not merely the speed of mathematical work but the conceptual organization of mathematics itself [7]. Commelin and Topaz show that deliberate abstraction boundaries can already make extremely complicated mathematical developments tractable [3]. Carette, Farmer, Kohlhase, and Rabe describe a related “one-brain barrier”: modern mathematical developments may exceed what a single human can survey [2].

RIT begins from these observations but asks a narrower question:

At what level should humans work when the mathematical process below that level has become too large to remain a reasonable human working surface?

The answer proposed here is **representation inversion**.

2 The status of RIT

RIT is not a new foundational system. It does not compete with set theory, type theory, Lean, category-theoretic foundations, little theories, MMT, formal libraries, or mathematical knowledge-management systems.

It also does not claim that mathematics has only now discovered abstraction. A theorem already allows later mathematicians to use a result without replaying its proof. A concept such as a Hilbert space allows a large collection of properties and operations to be handled as one cognitive object. A mathematician routinely relies on results proved by specialists in fields they do not personally reconstruct. Formal proof assistants already distinguish large processes of proof production from smaller mechanisms of proof checking. Large formalization projects use specifications, module boundaries, dependency graphs, and interfaces to prevent every participant from having to understand every internal detail simultaneously.

These are all predecessors of the phenomenon RIT describes.

The new claim concerns their *organizational role under machine-scale production*. The architecture is not new. What is new is the pressure for such architecture to become a normal condition of mathematical work.

RIT is therefore a philosophy of mathematical practice. Its subject is not primarily what mathematical objects exist. Its subject is the level at which mathematical agency, attention, responsibility, and judgment should operate.

3 Representation inversion

Suppose a mathematical region contains definitions, lemmas, proofs, computational experiments, alternative formulations, failed approaches, dependencies, representations, and a

large neighborhood of derivable consequences. Suppose further that a machine can search or expand this region much more cheaply than a human can comprehend that expansion.

There are two bad responses.

The first is to require humans to inspect progressively larger derivational traces. This destroys much of the advantage of automation.

The second is to accept machine-produced conclusions merely because a machine produced them. This abandons human responsibility for mathematical meaning, assumptions, relevance, interpretation, and integration into existing knowledge.

Representation inversion proposes a third response. Instead of making humans operate directly on the expanded mathematical region, the project constructs a smaller **working representation** through which that region can be used.

The interior is not discarded. It remains available in formal libraries, proofs, papers, computational records, databases, derivational histories, explanations, and machine-searchable structures. What changes is what must remain actively present in human work.

Representation Inversion Principle. When the expansion of a mathematical representation becomes substantially cheaper to generate or search than to comprehend directly, mathematical practice should seek a higher-order working representation that preserves the relevant mathematical capacity while reducing the amount that must remain cognitively active.

The important form of compression is therefore not information-theoretic compression. It is **operational and cognitive compression with recoverable access to the underlying mathematics**.

Representation inversion does not mean forgetting. It means changing the granularity at which mathematics is normally manipulated.

4 Interfaces are not walls

A natural objection is that hiding mathematical interiors produces opacity. That danger is real. RIT therefore does not treat an interface as an irreversible black box.

A useful interface must preserve the possibility of reopening the mathematical interior when the project requires it. This reopening need not be all-or-nothing. A user may first inspect assumptions and provenance, then dependency structure, then intermediate constructions, then explanatory material, and only in exceptional cases reconstruct a full derivational history.

Auditability is therefore naturally graded. The relevant principle is

operational compression + recoverable, graded access.

The interface should expose enough mathematical structure for the present task while preserving routes into deeper levels when those structures become relevant.

RIT therefore localizes opacity rather than pretending to eliminate it. Mathematics already depends on results that individual mathematicians do not fully reconstruct. It already contains enormous proofs, specialist knowledge, computational arguments, formal developments, and distributed expertise.

The question is not whether dependence can be eliminated. The question is whether its location can be made explicit.

A well-designed architecture should allow a project to ask:

- Which mathematical region is currently being delegated?
- What may safely be assumed from it?
- What assumptions cross its boundary?
- What provenance accompanies those assumptions?
- What remains unresolved?
- What event requires the region to be reopened?
- Who is responsible for judging whether the current interface is adequate?

RIT therefore delegates expansion. It does not delegate responsibility.

5 Kernels: mathematical regions relative to a project

To make the philosophy operational, RIT introduces a small architectural vocabulary. A mathematical region may be represented schematically as

$$K = (G, I, D, \partial),$$

where:

- G is the ground structure: assumptions, definitions, conventions, and primitive objects;
- I is the interface through which the present project interacts with the region;
- D is the larger interior that does not need to remain continuously active;
- ∂ is the unresolved frontier relevant to the present project.

These are organizational roles rather than new mathematical objects.

The distinction that matters most is between a **Theorem Kernel** and a **Research Kernel**. A region functions as a Theorem Kernel when its unresolved internal structure is not part of the current research problem. The project can therefore use the region mainly through its interface. A region functions as a Research Kernel when some unresolved part of that region matters directly to the current problem.

This distinction is project-relative. The same mathematical field may function as mature infrastructure in one project and as an active research object in another. RIT therefore does not divide mathematics permanently into settled and unsettled domains. It asks which parts of mathematics must remain open *for this problem, at this time, for this purpose*.

This project-relativity is essential because no abstraction boundary is universally sufficient. What can safely remain hidden depends on what one is trying to do.

6 The connector and the frontier

The most important human work in RIT occurs at the point where mathematical regions are made to interact in a way that has not yet become routine. RIT calls the mathematical product that establishes such an interaction a **connector**.

A connector may be a theorem, construction, reduction, equivalence, representation change, argument, or larger piece of research. Schematically,

$$K_0 \xleftrightarrow{\rho} K_1.$$

The symbol ρ is not merely a software adapter. It represents mathematical content that the current project has not yet delegated to established infrastructure.

The connector may be the central theorem of the paper. It may show that an object in one field admits a useful representation in another. It may transfer an estimate. It may identify two previously separate invariants. It may expose a hidden common structure.

It may also fail. A failed connector can reveal a missing hypothesis, an inadequate abstraction boundary, a false analogy, or an entirely new mathematical region.

The frontier is therefore where the present project concentrates responsibility. An AI system may propose connectors. It may search for them, formalize them, improve them, or eventually become better than humans at discovering many of them. RIT does not depend on a permanent division between what machines can do and what humans can do.

Its division is instead about **responsibility under conditions of unequal scale**. At the frontier, humans remain responsible for questions such as:

- Does this connection express the intended mathematics?
- Are the relevant assumptions visible?
- Is this result significant?
- Does the representation preserve what matters?
- Should this connection become part of shared mathematical infrastructure?

The philosophical claim is therefore not that machines calculate while humans understand. It is that **machine-scale production changes where scarce human responsibility should be concentrated**.

7 Stabilization, reopening, and recursive inversion

A connector does not remain frontier mathematics forever. If it becomes reliable, reusable, understood well enough for its intended purposes, and repeatedly used without requiring its reconstruction, it can become part of the mathematical interior.

The lifecycle is approximately

$$\text{interior} \rightarrow \text{interface} \rightarrow \text{frontier} \rightarrow \text{connector} \rightarrow \text{stabilization} \rightarrow \text{interior}.$$

This is ordinary mathematical development viewed through a different organizational lens. Yesterday's research theorem becomes tomorrow's lemma. A difficult construction becomes a standard tool. An entire theory becomes prerequisite infrastructure.

The reverse process is equally important. A previously stable mathematical region can be reopened when a new problem makes hidden structure relevant. RIT therefore does not freeze mathematics into black boxes. Its abstractions are provisional and indexed by use.

This produces a recursive principle. Suppose interfaces themselves proliferate until they become too numerous or too complicated for humans to manipulate effectively. Then the interface layer has become the new bottleneck. The appropriate response is another inversion.

Recursive Inversion Principle. When the current working representation becomes the bottleneck of mathematical work, search for a higher-order representation that preserves the relevant use while reducing the active surface.

This principle is important because machine capabilities need not stop at the current frontier. If machines eventually generate not merely millions of proof steps but thousands of useful interfaces, theories, or connectors, the human working level may again need to move.

RIT therefore describes no final representational layer. It describes a response to recurring asymmetry between the scale of available mathematical production and the scale of human mathematical attention.

8 Why the problem is already concrete

The philosophical issue does not require waiting for a universally accepted machine-generated solution to a major open problem.

In September 2026, OpenAI reported a proposed solution of the Navier–Stokes Millennium Prize Problem together with a Lean formalization. The report describes approximately 2.7 million agent messages and roughly 130 billion output tokens in the Navier–Stokes effort alone [11].

The eventual mathematical status of the proposed proof is a separate question. A Millennium Prize claim requires publication, time, scrutiny, and broad community acceptance [12]. That process has not occurred.

But RIT does not require the result to have completed that process. The organizational fact is already visible. No mathematical community can reasonably treat millions of agent interactions and hundreds of billions of generated tokens as the normal surface through which the result is understood.

If the underlying proof is correct, it must ultimately be transformed into mathematical objects that humans can evaluate: central constructions, assumptions, mechanisms, dependency structures, formal certificates, conceptual explanations, connections with existing theory, and questions for further work.

If the proof is wrong, essentially the same representational problem arises: mathematicians need ways to locate where the argument fails without replaying an enormous machine search.

The problem therefore precedes final acceptance. Machine-scale derivation has made the issue immediate.

9 A division of mathematical labour

The consequence of RIT is not that mathematicians stop proving theorems. It is that the default allocation of effort changes when derivational production becomes abundant.

Different activities can occur at different representational levels:

machine systems → large-scale search and derivation,
 formal systems → certificate checking,
 mathematicians → representation, meaning, frontier judgment,
 specialists and communities → audit, interpretation, stabilization.

These are roles rather than permanent professions. The same person or system may occupy several of them. Future AI systems may occupy more of them than current systems do.

The philosophical claim does not depend on drawing an eternal border around human cognition. It depends on recognizing that when one representational layer becomes cheap to generate but expensive for humans to assimilate, scarce human effort should move toward the layers at which judgment remains consequential.

This is an economic claim in the broad sense. Mathematical attention is finite. Specialist time is finite. Community review is finite. Formalization effort, computation, maintenance, and explanation all have costs. RIT proposes that representation itself should respond to those changing scarcities.

10 The broader RIT research programme

The present paper deliberately stops at the philosophical stance. RIT is intended as a division-of-labour research programme with several distinct but interacting components.

10.1 Philosophical stance: the present paper

The philosophical stance asks:

Why should the level of human mathematical work change under machine-scale derivation?

It defines representation inversion, project-relative kernels, interfaces, connectors, reopening, stabilization, and recursive inversion as concepts for organizing mathematical agency. Its claim is normative and organizational. It does not attempt to prove that a particular interface is mathematically optimal.

10.2 Epistemic and economic stance

A second line of work must ask whether representation inversion actually improves mathematical practice. Its questions are concrete:

- How much human attention does an interface save?

- Which assumptions remain visible?
- When do humans need to reopen an interior?
- How often do interfaces conceal relevant information?
- How much specialist work is required to audit a connector?
- How does trust change when mathematical derivation is machine-generated?
- When does kernelization save labour, and when does maintaining the interface cost more than it saves?

This part of the programme belongs in real mathematical case studies. Its task is not to justify the philosophical possibility of inversion but to determine where inversion is epistemically and economically successful.

10.3 Mathematical stance

A third line of work must make the architecture mathematically concrete. Its question is:

How can existing mathematics be represented in the language of kernels, interfaces, frontiers, and connectors without replacing the mathematical structures already in use?

Existing formal systems already provide many candidate mechanisms: theorem declarations, modules, theory morphisms, namespaces, dependency graphs, specifications, type-theoretic structures, category-theoretic constructions, and formal libraries.

The mathematical programme should therefore embed RIT into existing mathematics rather than invent a universal replacement language. Different mathematical fields may instantiate the architecture differently. Determining a good interface is itself a mathematical problem.

10.4 Complexity-theoretic stance

A fourth line of work concerns the reason representation inversion repeatedly becomes necessary. The philosophical paper begins with the inequality

$$\text{machine derivational capacity} > \text{human assimilative capacity}.$$

A complexity-theoretic treatment should investigate this relation more precisely. It should ask when one representational layer becomes too expensive for human use, what quantities characterize that transition, and when the resulting higher-order representation itself becomes the next bottleneck.

This work supplies the deeper theory behind **recursive inversion**. Its purpose is not merely to explain the present AI transition. It should explain how mathematical organization responds to future crises of scale as well.

11 What RIT claims – and what it does not

RIT does not claim that abstraction is new. It does not claim that mathematics has never before exceeded individual human understanding. It does not claim that every mathematical

region should become a kernel. It does not claim that a small interface automatically preserves understanding. It does not claim that machines will never work at mathematical frontiers. It does not claim that formal verification eliminates semantic judgment. And it does not claim that opacity can be removed from sufficiently large mathematical systems.

Its narrower claim is this:

Machine-scale derivation changes the ratio between mathematical production and human assimilation strongly enough that the representation on which human mathematical labour operates may itself have to change.

RIT calls that change **representation inversion**.

The practical success of a particular inversion remains a mathematical, epistemic, and economic question. But those questions arise only after the philosophical possibility is recognized.

12 Conclusion

Mathematics has always depended on compression. Definitions compress patterns. Concepts compress families of ideas. Theorems compress arguments into reusable results. Specialization compresses entire fields into trusted dependencies. Formal verification separates enormous derivational processes from smaller trusted checking mechanisms.

Machine reasoning pushes this historical process to a new scale.

When a mathematical system can generate or search more structure than humans can plausibly assimilate, requiring humans to remain at the same representational level ceases to be a sensible organization of mathematical work.

The correct response is not to make humans consume machine-scale derivation faster. It is to change what humans must keep active.

Stable regions can become project-relative infrastructure. Interfaces can expose the mathematics required for current use. Interiors can remain recoverable at graded levels of audit. Unresolved crossings can become explicit frontiers. Connectors can concentrate mathematical responsibility where new meaning is being established. Successful connectors can later become infrastructure themselves.

If the new representation eventually becomes too large, the process repeats.

The purpose of RIT is therefore not to remove human beings from mathematics. It is to preserve meaningful human mathematical agency when the scale at which mathematics can be produced no longer matches the scale at which humans can directly comprehend its production.

When machine capacity overwhelms one level of mathematical representation, humans need not compete at that level.

They change the level at which mathematics is done.

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